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B.Tech. Degree VIII Semester (Supplementary) Examination in Mechanical Engineering, May 2003

ME 802 CAD/CAM II (1995 & 1998 Admissions)

Time: 3 Hours

Maximum Marks: 100

- I. (a) How CNC differ from DNC? Make a comparison. (10)
- (b) Detail the "Detroit type of automation". (10)
- OR**
- II. (a) Distinguish between a pallet and fixture. (5)
- (b) With neat sketches explain absolute and incremental types of linear and angular encoders. (15)
- III. (a) Explain the APT programming language. (15)
- (b) What is meant by "canned cycle"? Give two examples. (5)
- OR**
- IV. (a) Why the term "Artificial" for the concept AI and how it differ from natural intelligence? Give suitable example to complement your argument. (6)
- (b) How do you specify an industrial robot? (10)
- (c) What is meant by "Degree Of Freedom (DOF)" for a robot? Make a simple sketch of a robot having 5DOF. (4)
- V. (a) With necessary sketches, explain the types and the working of Re-circulating ball screws. (10)
- (b) Discuss the measuring systems used in Numerically Controlled machines. (10)
- OR**
- VI. Explain the special design features to match machine tools to Numerical Control. (20)
- VII. (a) Explain different types of automatic tool loading mechanisms in machining centers, with comparative features. (10)
- (b) Discuss the principles involved in fixture design. (10)
- OR**
- VIII. (a) How to design fixtures for different applications? (10)
- (b) Explain the concept "Single Minute Exchange of Dies (SMED)" with an example. (10)
- IX. Explain the types, applications and specialities of different types of dies for sheet metal operations, with sketches. (20)
- OR**
- (b) Discuss about the design of die sets for different sheet metal operations. (15)
- (b) Write the difference between punching and blanking operations. (5)

